Example 1:

Calculate the hole and electron densities in a piece of p-type silicon that has been doped with $5\ x\ 10^{16}$ acceptor atoms per cm^3. $n_i = 1.5\ x\ 10^{10}\ cm^{-3}$ (at room temperature).

Solution:

P-type ; $p \gg n$

$n.p = {n_i}^2$ $\quad N_A = 5\ x\ 10^{16}$ $\quad p = N_A = 5\ x\ 10^{16}\ cm^{-3}$

$$n = \frac{{n_i}^2}{p} = \frac{(1.5\ x\ 10^{10}\ cm^{-3})^2}{5\ x\ 10^{16}\ cm^{-3}}$$

$$= 4.5\ x\ 10^3\ electrons\ per\ cm^3$$

NB:

$p \gg n_i$ and $n \ll n_i$ in a p-type material. The more holes you put in the less electrons you have and vice versa

Carrier Movement

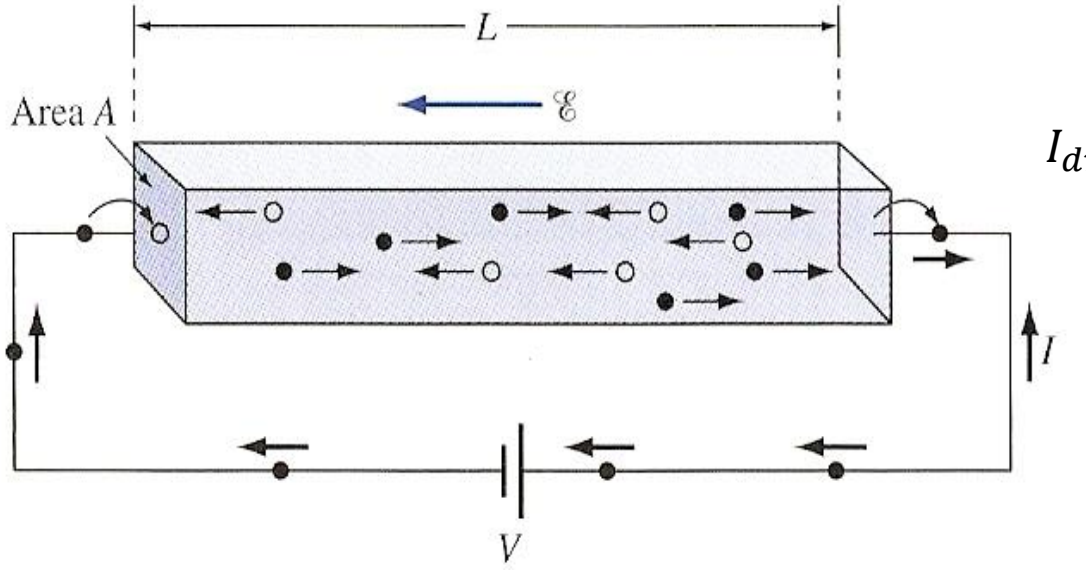
- There are two mechanisms by which holes and free electrons move through a silicon crystal.
- Drift--- The carrier motion is generated by the electrical field across a piece of silicon. This motion will produce drift current.
- Diffusion--- The carrier motion is generated by the different concentration of carrier in a piece of silicon. The diffused motion, usually carriers diffuse from high concentration to low concentration, will give rise to diffusion current.

Carrier Mobility

- Mobility is a measure of how easily a carrier moves in particular material.
- It is normally expressed in centimetres squared per volt-second (cm^2/Vs).
- Carrier mobility, $\mu = \frac{v}{E}$ and it is the drift velocity per unit field.
- Drift velocity is the average velocity with which the charge carriers drift in a definite direction under the influence of an applied electric field.
- The unit of μ is $\frac{cm/s}{V/cm} = cm^2/Vs$

Drift Current

- Drift current is produced by the motion of electrons and holes under an electric field.
- The net current is always in the direction of applied electric field
- The current due to both charges move in the same direction.



$$I_{drift} = I_{n(drift)} + I_{p(drift)}$$

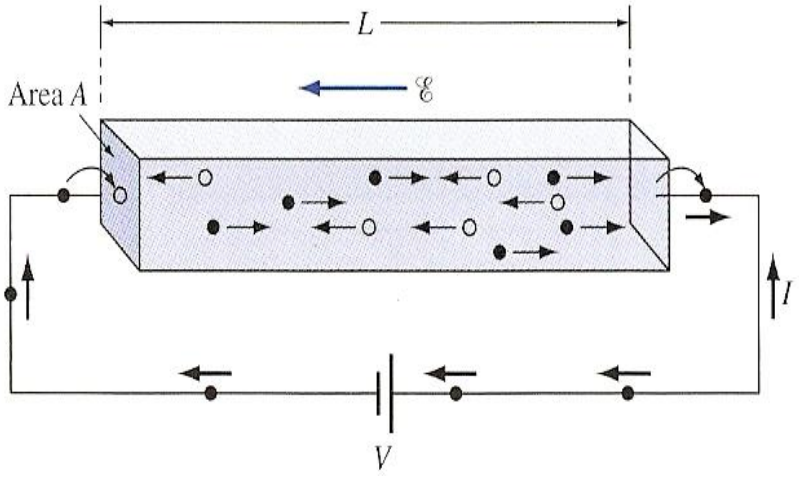
Current Density

- In a semiconductor, it is often more useful to talk about current density $\boldsymbol{J}$, which is the amount of charge crossing a plane unit area per unit time.
- That is : $$\mathbf{J} = \frac{\boldsymbol{I}}{\boldsymbol{A}}$$
- For a semiconductor with an applied electric field, the total drift current is

$$\boldsymbol{J_{drift}} = \boldsymbol{J_{n(drift)}} + \boldsymbol{J_{p(drift)}}$$

where $J_{n(drift)}$ is the current due to electrons

$J_{p(drift)}$ is the current due to holes



Conductivity

- When an electric field E is applied to a semiconductor, the free electrons drift with average velocity $\boldsymbol{v_e = -\mu_e E}$, where μ_e is the electron mobility.
- The holes drift with a velocity $\boldsymbol{v_h = +\mu_h E}$, where μ_h is the hole mobility.
- Although the free electrons and holes drift in opposite directions, the current densities add because the charge polarities are opposite.
- The electron charge density is $\boldsymbol{\rho_e = -nq}$ and the hole charge density is $\boldsymbol{\rho_h = +pq}$, where n and p represent the electron and hole concentrations respectively.
- The total conduction current density can be written as

$$\boldsymbol{J = \rho_e v_e + \rho_h v_h = (n\mu_e + p\mu_p)qE = \sigma E}$$

- This equation defines the conductivity σ of the semiconductor. It is given by

$$\boldsymbol{\sigma = (n\mu_e + p\mu_p)q}$$

Conductivity

- Alternatively, the conductivity can be obtained from the dimensions of the semiconductor as follows.
- From ohm's law, the resistance $\boldsymbol{R}$ of a uniform sample of length $\boldsymbol{L}$ and cross-sectional area $\boldsymbol{A}$ as illustrated in the semiconductor block below, is

$$\boldsymbol{R = \frac{V}{I} = \frac{\rho L}{A}}$$

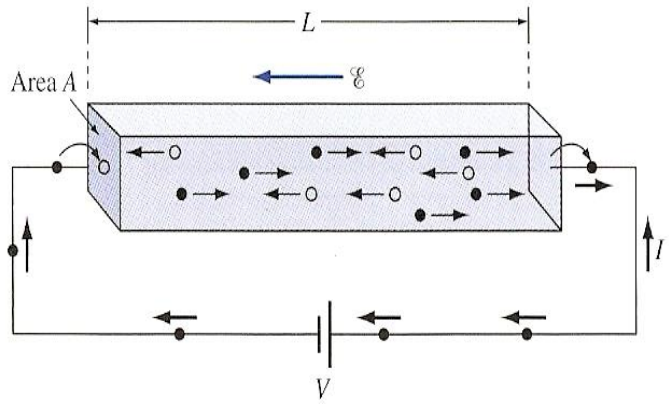
- Since the semiconductor is uniform, $\boldsymbol{V = EL}$ and using $\boldsymbol{I = JA}$,

$$\Rightarrow \boldsymbol{\frac{EL}{JA} = \frac{\rho L}{A}}$$

$$\Rightarrow \boldsymbol{J_{drift} = \frac{E}{\rho} = \sigma E}$$

where $\boldsymbol{\sigma = \frac{1}{\rho}}$ is the conductivity

E is the electric field and ρ is the resistivity



Example 2

- A rod of intrinsic silicon is 1 cm long and has a diameter of 1 mm. At room temperature, the intrinsic concentration in the silicon is $n_i = 1.5\ x\ 10^{16}\ per\ m^3$. The electron and hole mobilities are $\mu_e = 0.13\ m^2V^{-1}s^{-1}$ and $\mu_h = 0.05\ m^2V^{-1}s^{-1}$. Calculate the conductivity of the silicon and the resistance R of the rod.

Solution:

- The conductivity is calculated as follows:

$$\sigma = n_i(\mu_e + \mu_h)q$$
$$= 1.5 \text{ x } 10^{16} \text{ x } (0.13 + 0.05) \text{ x } 1.602 \text{ x } 10^{-19}$$
$$= 4.33\ x\ 10^{-4}\ S/m$$

- Resistance is calculated as:

$$R = \frac{l}{\sigma A} = \frac{0.01}{4.33 \text{ x } 10^{-4} \text{ x } \pi(0.5 \text{ x } 10^{-3})^2} = 29.4\ \text{M}\Omega$$

Temperature sensitivity

- In both types of extrinsic semiconductor virtually all available charge carries are freed from their parent atoms at room temperature. Temperature variations thus make little difference to the conductivity σ.
- For intrinsic conductivity the number of carriers, and thus σ, increases rapidly with temperature.
- For both extrinsic and intrinsic mechanisms the conductivity is zero at T=0 K

Diffusion Current

- The directional movement of charge carriers due to their concentration gradient produces a component of current known as *diffusion current*.
- It is encountered only in semiconductors and is normally absent in conductors
- With no applied voltage, if the number of charge carriers (either holes or electrons) in one region of a semiconductor is less compared to the rest of the region, then there exists a concentration gradient.
- As a result, the carriers tend to move gradually or diffuse from the region of higher concentration to the region of lower concentration.
- This process is called diffusion and electric current produced due to this process is called diffusion current.
- This process continues until all the carriers are evenly distributed through the material. Hence when there is no applied voltage, the net diffusion current will be zero

Diffusion Current

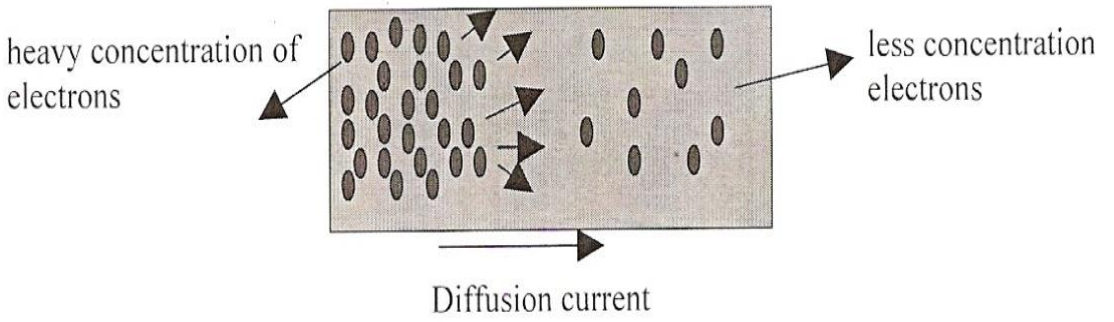
- In general, the hole diffusion current density is given by

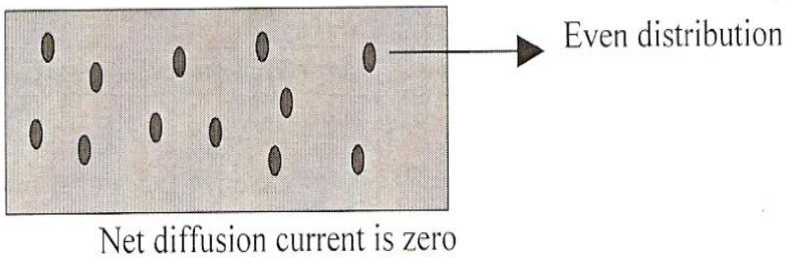
$$J_p = -qD_h\nabla_p$$

Where D_h is the hole diffusion constant and ∇_p is the gradient of p

- Similarly, the electron diffusion current density is given by

$$J_n = +qD_e\nabla_n$$





Total Current

- In general, the total current in a semiconductor is given by the sum of the electron and hole drift currents and the electron and hole diffusion currents.

$$J = \sigma E + q(D_e \nabla_n - D_h \nabla_p)$$

Carrier Lifetime /Mean Lifetime

- Because hole-electron pairs are continually created by thermal agitation of a semiconductor lattice, it might seem that the number of holes and free electrons would continually increase with time.
- This does not happen because free electrons are continually recombining with holes.
- At any temperature, a stable state is reached when the creation rate of hole-electron pairs is equal to the recombination rate.
- The mean lifetime τ_n of a free electron is the average time that the electron exists in the free state before recombination.
- The mean lifetime τ_p for the hole is defined similarly. In the intrinsic semiconductor, τ_n is equal to τ_p because the number of free electrons must be equal to the number of holes.

Terminology

donor: impurity atom that increases *n*

acceptor: impurity atom that increases *p*

N-type material: contains more electrons than holes

P-type material: contains more holes than electrons

majority carrier: the most abundant carrier

minority carrier: the least abundant carrier

intrinsic semiconductor: $n = p = n_i$

extrinsic semiconductor: doped semiconductor

In Summary

- In its pure state, semiconductor material is an excellent insulator.
- The commonly used semiconductor material is silicon.
- Semiconductor materials can be doped with other atoms to add or subtract electrons.
- An N-type semiconductor material has extra electrons.
- In an N-type semiconductor, conduction is mainly due to electrons (negative charges). Positive charges (holes) are the minority carriers.
- A P-type semiconductor material has a shortage of electrons with vacancies called holes.
- In a P-type semiconductor, conduction is mainly due to holes (positive charges). Negative charges (electrons) are the minority carriers.
- The heavier the doping, the greater the conductivity or the lower the resistance.
- By controlling the doping of silicon the semiconductor material can be made as conductive as desired.

…that's all folks…
…thanks for your time…

